Mohammad Raihanul Bashar, PhD

mbraihan.github.io mohammadraihanul.bashar@mail.concordia.ca Google Scholar

Research Interests

Human-Computer Interaction · Extended Reality (XR) · Human-AI Interaction · Multimodal Interaction · Adaptive User Interfaces · Generative User Interfaces · Human-Centered AI

Education

01/2024 – 08/2026 **Ph.D. in Computer Science**
Department of Computer Science & Software Engineering, Concordia University
Advisor: Anil Ufuk Batmaz
Thesis: Constraint-Aware Selection in Extended Reality
Committee: Mohammad Khalad Hasan, Mahdi Mirhoseini, Marta Kersten-Oertel, Lata Narayanan

09/2023 – 12/2023 **MSc in Computer Science**
Department of Computer Science & Software Engineering, Concordia University
Fast-track to PhD
Advisor: Anil Ufuk Batmaz

01/2013 – 08/2017 **BSc in Computer Science**
Department of Computer Science, Independent University, Bangladesh
Advisor: Prof. M. Ashraful Amin
Thesis: Effect of EMG Artifacts on Video Category Classification from EEG

Publications

Papers with future publication dates are accepted to appear in their respective venues.

Book Chapters

[B.1] **Computational Intelligence for Pattern Recognition in EEG Signals**
Aunnoy K Mutasim, Rayhan Sardar Tipu, Mohammad Raihanul Bashar, Md Kafiul Islam, M. Ashraful Amin
Computational Intelligence for Pattern Recognition, Springer-Verlag

Peer-Reviewed Hybrid Conference-Journal Articles

[J.5] **Evaluating the Vergence–Accommodation Conflict in Gaze-Based 3D Target Selection**
Mohammad Raihanul Bashar, Mohammadreza Amini, Aunnoy K Mutasim, Mayra Donaji Barrera Machuca, Wolfgang Stuerzlinger, and Anil Ufuk Batmaz
TVCG: IEEE Transactions on Visualization and Computer Graphics (IEEE ISMAR' 26)
Best Paper Award Honorable Mention

[J.4] **An Early Warning System Based on Visual Feedback for Light-Based Hand Tracking Failures in VR Head-Mounted Displays**
Mohammad Raihanul Bashar, Anil Ufuk Batmaz
TVCG: IEEE Transactions on Visualization and Computer Graphics (IEEE VR' 25)

Peer-Reviewed Journal Articles

[U.4] **Gamification as Formative Feedback in Virtual Reality: Ultrasound-Guided Catheter Insertion Training for Novices and Experts**
Mohammad Raihanul Bashar, Alejandro Olivares Hernandez, Yahia Zine, Anil Ufuk Batmaz
TVCG: IEEE Transactions on Visualization and Computer Graphics

[J.3] **The Effect of Visual Depth on the Vergence-Accommodation Conflict on 3D Selection Performance within Virtual Reality Headsets**
Mohammad Raihanul Bashar, Mayra Donaji Barrera Machuca, Wolfgang Stuerzlinger, Anil Ufuk Batmaz
The Visual Computer, Springer-Verlag

[J.2] **EEG-Based Preference Classification for Neuromarketing Application**
Injamamul Haque Sourov, Faiyaz Alvi Ahmed, Md. Tawhid Islam Opu, Aunnoy K Mutasim, M. Raihanul Bashar, Rayhan Sardar Tipu, M Ashraful Amin, and Md Kafiul Islam
Computational Intelligence and Neuroscience

[J.1] **Use of spontaneous blinking for application in human authentication**
Amir Jalilifard, Dehua Chen, Aunnoy K Mutasim, M. Raihanul Bashar, Rayhan Sardar Tipu, Md. Ahsan-Ul Kabir Shawon, Nazmus Sakib, M Ashraful Amin, and Md Kafiul Islam
Engineering Science and Technology, an International Journal, Elsevier

Peer-Reviewed Conference Proceedings

[U.3] **Multi-Selection in XR**
CHI 2027: ACM CHI Conference on Human Factors in Computing Systems

[U.2] **Short Dwell Threshold Errors**
CHI 2027: ACM CHI Conference on Human Factors in Computing Systems

[U.1] **Generative User Interfaces**
CHI 2027: ACM CHI Conference on Human Factors in Computing Systems

[C.9] **Crossing Realities Has a Cost: Evaluating AR Second Screens for AI-Assisted Desktop Authoring**
Mucahit Gemici, Mohammad Raihanul Bashar, Wolfgang Stuerzlinger, Vrushank Phadnis, Anil Ufuk Batmaz
SUI 2026: ACM Symposium on Spatial User Interaction

[C.8] **Eyes on Many: Evaluating Gaze, Hand, and Voice for Multi-Object Selection in Extended Reality**
Mohammad Raihanul Bashar, Aunnoy K Mutasim, Ken Pfeuffer, Anil Ufuk Batmaz
CHI 2026: ACM CHI Conference on Human Factors in Computing Systems

[C.7] **There Is More to Dwell Than Meets the Eye: Toward Better Gaze-Based Text Entry Systems With Multi-Threshold Dwell**
Aunnoy K Mutasim, Mohammad Raihanul Bashar, Christof Lutteroth, Anil Ufuk Batmaz, Wolfgang Stuerzlinger
CHI 2025: ACM CHI Conference on Human Factors in Computing Systems

[C.6] **Virtual Task Environments Factors Explored in 3D Selection Studies**
Mohammad Raihanul Bashar, Anil Ufuk Batmaz
GI 2024: ACM Graphics Interface

[C.5] **Effect of Artefact Removal Techniques on EEG Signals for Video Category Classification**
Aunnoy K Mutasim, Mohammad Raihanul Bashar, Rayhan Sardar Tipu, Md Kafiul Islam, M. Ashraful Amin
ICPR 2018: International Conference on Pattern Recognition

[C.4] **Effect of EMG Artifacts on Video Category Classification from EEG**
Mohammad Raihanul Bashar, Rayhan Sardar Tipu, Aunnoy K Mutasim, M. Ashraful Amin
ICIEV 2018: International Conference on Informatics, Electronics & Vision

[C.3] **Video Category Classification Using Wireless EEG**
Aunnoy K Mutasim, Rayhan Sardar Tipu, Mohammad Raihanul Bashar, M. Ashraful Amin
BI 2017: International Conference on Brain Informatics

[C.2] **Implementation of Low Cost Stereo Humanoid Adaptive Vision for 3D positioning and Distance Measurement for Robotics Application with Self-calibration**
Abul Al Arabi, Rayhan Sardar Tipu, Mohammad Raihanul Bashar, Binoy Barman, Shama Ali Monica, M. Ashraful Amin
AMS 2017: Asia Modelling Symposium

[C.1] **2D Surface Mapping for Mine Detection via Wireless Networking**
Mohammad Raihanul Bashar, Abul Al Arabi, Rayhan Sardar Tipu, Md Tanvir Alam Sifat, Md Zobair Ibnalam, M. Ashraful Amin
ICCRE 2017: International Conference on Control and Robotics Engineering

Peer-Reviewed Conference Poster Abstracts, Doctoral Consortium, Workshop

[D.1] **Context-Aware Modality Adaptation for Robust Multimodal XR Interaction**
Mohammad Raihanul Bashar
ISMAR 2026 Doctoral Consortium

[A.2] **Effects of Visual Depth on Vergence–Accommodation Conflict in Gaze- and Controller-Based Selection in Virtual Reality**
Mohammad Raihanul Bashar, Mohammadreza Amini, Aunnoy K Mutasim, Mayra Donaji Barrera Machuca, Wolfgang Stuerzlinger, Anil Ufuk Batmaz
CHI 2026 Extended Abstract: ACM Conference on Human Factors in Computing Systems

[A.1] **Depth3DSketch: Freehand Sketching Out of Arm's Reach in Virtual Reality**
Mohammad Raihanul Bashar, Mohammadreza Amini, Wolfgang Stuerzlinger, Mine Sarac, Ken Pfeuffer, Mayra Donaji Barrera Machuca, Anil Ufuk Batmaz
CHI 2025 Extended Abstract: ACM Conference on Human Factors in Computing Systems

[W.1] **Subtask-Based Virtual Hand Visualization Method for Enhanced User Accuracy in Virtual Reality Environments**
Laurent Voisard, Amal Hatira, Mohammad Raihanul Bashar, Mucahit Gemici, Mine Sarac, Marta Kersten-Oertel, Anil Ufuk Batmaz
IEEE VR 2024 Workshop: Novel Input Devices and Interaction Techniques

Research Experience

12/2022 - *current*

EXiT Lab, Concordia University, Graduate Student Researcher. Montreal, QC. Advisor: Dr. Anil Ufuk Batmaz. Led research on constraint-aware, multimodal interaction in extended reality, spanning gaze, hand tracking, voice, controllers, and cross-reality interfaces. Designed and implemented five XR research testbeds and conducted six controlled studies with 140 participants, using quantitative, qualitative, and mixed-methods evaluation. Developed a depth-aware predictive model of 3D selection, a proactive warning system for hand-tracking degradation, and serial and parallel multi-object selection techniques. Led research activities including problem formulation, interaction design, Unity/OpenXR implementation, experiment design, statistical analysis, visualization, and manuscript preparation. Extended this work toward AI-assisted authoring and generative user interfaces.

01/2019 – 12/2021

AGenCy Lab, IUB, Independent University, Bangladesh, Research Assistant. Dhaka, BD. Advisor: Prof. M. Ashraful Amin. Develop a high-quality, standardized medicine image dataset with metadata, deploy a recognition model to enhance feature selection, and create a mobile application to assist visually impaired patients in identifying their prescribed pills. Work led to opensource code [O. 3].

01/2018 – 12/2019

Computer Vision & Cybernetics, BD, Independent University, Bangladesh, Research Assistant. Dhaka, BD. Advisor: Prof. M. Ashraful Amin. The goal was to compile a dataset of Bengali letters and numbers for Bangladeshi license plates and develop a custom ANPR system, successfully addressing the lack of open-source datasets and enabling accurate recognition of Bangladeshi license plates. Work led to opensource code [O. 2].

04/2016 – 01/2018

Computer Vision & Cybernetics, BD, Independent University, Bangladesh, Undergraduate Research Assistant. Dhaka, BD. Advisor: Prof. M. Ashraful Amin. With the vision of building a BCI-based video recommender system, we experimented with several state-of-the-art algorithms for each of the submodules (pre-processing, feature extraction, feature selection, and classification) of the Signal Processing module of a BCI system to find out what combination of algorithms best predicts what type of video a person is watching. Work led to publication at BI 2017 [C. 3], ICIEV 2017 [C. 4], ICPR 2018 [C. 5] and a Book Chapter [B. 1] and opensource code [O. 1].

Teaching

Instructor of Record

Winter 2026 — **SOEN 287: Web Programming**
Undergraduate, 113 students
Concordia University

Teaching Assistant

Fall 2026 — **COMP 248: Introduction to Programming**
Undergraduate, 27 students
Instructor: Dr. Malleswara Rao Talla, Concordia University

Summer 2026 — **SOEN 6011: Software Engineering Process**
Graduate, 32 students
Instructor: Dr. Pankaj Kamthan, Concordia University

Summer 2026 — **SOEN 6471: Advanced Software Architectures**
Graduate, 37 students
Instructor: Dr. Pankaj Kamthan, Concordia University

Summer 2026 — **COMP 248: Introduction to Programming**
Undergraduate, 25 students
Instructor: Dr. Pankaj Kamthan, Concordia University

Winter 2026 — **COMP 6741: Intelligent Systems**
Graduate, 42 students
Instructor: Dr. Pankaj Kamthan, Concordia University

Fall 2025 — **COMP 5201: Computer Organization and Assembly Language**
Graduate, 27 students
Instructor: Dr. David K. Probst, Concordia University

Fall 2025 — **COMP 472: Introduction to Artificial Intelligence**
Undergraduate, 38 students
Instructor: Dr. Kefaya Qaddoum, Concordia University

Fall 2023–2025 — **COMP 376: Introduction to Game Development**
Undergraduate
Instructor: Dr. Anil Ufuk Batmaz, Concordia University

Winter 2024 — **COMP 6331: Advanced Game Development**
Graduate
Instructor: Dr. Anil Ufuk Batmaz, Concordia University

Winter 2024 **SOEN 6751: Human-Computer Interaction**
Graduate, 103 students
Instructor: Dr. Anil Ufuk Batmaz, Concordia University

Fall 2018 **CSE 417: Data Mining and Warehouse**
Graduate, 57 students
Instructor: Dr. Moinul Zaber, Independent University, Bangladesh

Fall 2018 **CSE 420: Image Processing & Pattern Recognition**
Undergraduate, 33 students
Instructor: Prof. M. Ashraful Amin, Independent University, Bangladesh

Mentoring & Supervision

2024–2026 **Graduate and Undergraduate Research Mentor**
EXiT Lab, Concordia University
Mentored students on XR and HCI research, including study design, Unity development, experimental methodology, data analysis, and academic writing.

Industry Experience

Fall 2024 **Mitacs Accelerate Internship**, Research Intern. Remote. SonoNurse Inc., Concordia University, Montreal, QC.
Developed a gamified virtual reality (VR) training system for ultrasound-guided intravenous catheter insertion, incorporating real-time feedback and interaction design to enhance procedural learning. Led system design and implementation, resulting in a peer-reviewed publication [U.4].

05/2022 – 08/2023 **Senior Engineer, MLOps.** Quantigo AI, Dhaka, Bangladesh. Led the development and launch of two web applications, including a workforce management system, while providing customized ML-assisted data annotation solutions, developing an in-house video annotation platform, and establishing foundational architectures for current and future projects.

07/2021 – 03/2022 **Senior AI Engineer.** OPUS Technology Limited, Dhaka, Bangladesh. Developed an intelligent agent for CS:GO using offline RL, supervised the creation of an e-sports game prediction system, implemented a matchmaking system with AWS Gamelift, streamlined MLOps integration with cloud platforms, mentored the ML team on RL and clean coding practices, and maintained the company's largest project while contributing reusable modular-helper classes.

Summer 2021 **Machine Learning Engineer.** Expert Consortium Limited, Dhaka, Bangladesh. Developed an early shoplifting detection system prototype.

11/2020 – 03/2021 **Software Engineer.** Maxis Systems Limited, Dhaka, Bangladesh. Integrated a DevOps pipeline as the development platform, developed scalable and maintainable microservice solutions, and built a platform for business analytics leveraging ML and Data Science.

Awards and Scholarships

2026 **GCS Conference Travel Support, CAD 500**, Gina Cody School of Engineering and Computer Science, Concordia University

2025 **Concordia Conference and Exposition Allowance, CAD 1000**, School of Graduate Studies, Concordia University

2025 **Concordia Conference and Exposition Allowance, CAD 1000**, School of Graduate Studies, Concordia University

2025 **Mitacs Accelerate Internship, CAD 15000**, Mitacs, SonoNurse Inc.

2024 **Concordia Conference and Exposition Allowance, CAD 750**, School of Graduate Studies, Concordia University

2024 **FRS Scholarship for PhD Student, CAD 22,000/year**, Concordia University

2024 **Concordia International Tuition Award of Excellence, CAD 44,893**, Concordia University

2023 **FRS Scholarship for MSc Student, CAD 24,000/year**, Concordia University

2023 **Concordia Merit Scholarship, CAD 5,000**, Concordia University

2017 **SECS Award**, Independent University, Bangladesh

Service

Service items with future dates are confirmed commitments.

Leadership & Organizing Roles

2026 **Session Chair (Driving Innovation)**, ACM CHI 2026

2026 **Poster Associate Chair (AC)**, ACM CHI 2026, ACM SUI 2026

2026 **Workshop Co-Organizer**, NIDIT @ IEEE VR 2026

2026 **Student Volunteer**, IEEE ISMAR 2026

2025 **Student Volunteer Chair**, ACM SUI 2025

2025 **Student Volunteer**, IEEE VR 2025

Program Committee

2025–2027 **Program Committee Member**
IEEE VR 2027
ACM IUI 2026–2027
ACM SUI 2026
ACM ISS 2026
ACM MUM 2025

Reviewing

2024–2027 **Reviewer**
ACM CHI 2024, 2026–2027 *(Special Recognition for Outstanding Reviews, 2026)*
IEEE VR 2025, 2027
IEEE ISMAR 2024–2026
ACM UIST 2025–2026
IEEE TVCG 2025–2026
ACM MobileHCI 2024, 2026
ACM ICMI 2026
ACM SUI 2024, 2026
ACM VRST 2025–2026
ACM CHI LBW 2025

Community & Outreach

2024–2026 **Presenter**
Girls Summer Engineering and Technology Program (GirlSET), Concordia University

Skills

I have expertise in developing interactive XR and AI systems, as well as full-stack web development, encompassing both frontend and backend. Additionally, I have experience in designing, training, and evaluating deep learning models. I am familiar with the following programming languages, frameworks, and research methodologies.

Programming Languages: Python, C++, C#, HTML/CSS/JavaScript

Frameworks: OpenXR, PyTorch, React, Scikit-Learn, Unity, Unreal

Research Methodologies: User Studies (quantitative/qualitative), DL experiments.

References

- **Anil Ufuk Batmaz**
  Assistant Professor
  Gina Cody School of Engineering and Computer Science
  Concordia University
  ufuk.batmaz@concordia.ca

- **Wolfgang Stuerzlinger**
  Professor
  School of Interactive Arts + Technology
  Simon Fraser University
  w.s@sfu.ca

- **Mayra Donaji Barrera Machuca**
  Associate Professor
  Faculty of Science
  University of Calgary
  mbarrera@ucalgary.ca